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TESS Photometric Variability of Young Brown Dwarfs in the Taurus Star-forming Region

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Abstract

We present a comprehensive analysis of TESS high-quality light curves from sectors 43 and 44 of a few samples of young (∼2–3 Myr) brown dwarfs in the Taurus molecular cloud. They are well characterized and bona fide members of Taurus. We aim to search for the fast rotations of brown dwarfs and to picturize their dynamic atmosphere and surface features. Out of 11 young BDs, we found that 72% are periodic, in the period range of 1–7 days; among them, three BDs have periods <1.5 day and the period of one object is estimated for the first time. The sinusoidal periodic variations are related to a large spot or group of small spots corotating with the objects. Interestingly, we have detected four flare events in three young BDs, with one object, MHO 4, showing two flares in two different sectors. From the flared light curves, we have estimated the total bolometric flared energy in a range of 10^{35}–10^{36} erg, which is close to the superflare energy range ($\geqslant 10^{34}$ erg). To produce such kinds of superflare events, we have calculated the required magnetic field strength, which comes out at the order of a few kilogauss. Such superflares have a strong effect on the habitability of planets around M dwarfs.

Unified Astronomy Thesaurus concepts: Variable stars (1761); Stellar flares (1603); Stellar activity (1580); M dwarf stars (982); Starspots (1572); Stellar rotation (1629); Pre-main sequence stars (1290)

1. Introduction

Brown dwarfs (BDs) are traditionally defined as substellar objects with enough mass to sustain deuterium burning inside their core, but not enough for hydrogen burning (Saumon et al. 1996). They are intermediate objects between stars and planets in terms of mass, with a mass limit from 13 to 80 Jupiter masses (M_J). The radius is typically observed to be 0.7–1.4 Jupiter radii (R_J) for solar luminosity (Csizmadia 2016; Carmichael et al. 2019). According to the models of BD structure and evolution, the radii and luminosity of these objects decrease with object age as their interior cools down (Baraffe et al. 2003; Saumon & Marley 2008), and due to relatively low surface temperature, most of the radiation radiates in the IR portion of the spectrum. As these objects are fully convective with low temperatures, the surface features and atmospheres of BDs show several complexities that are generally not present in the atmosphere of stars (Kirkpatrick 2005).

Research into the photometric variability of BDs has received much attention in recent years, to understand these young objects' dynamic atmospheric properties and surface features (Zapatero Osorio et al. 2003; Caballero et al. 2004). Photometric variability refers to a change in a celestial object's brightness over time. BDs show various types of photometric variability, including periodic and aperiodic variations and short-duration flares. The periodic variability of BDs mainly arises due to the rotational modulation of active "starspot" regions in the photosphere, which causes changes in the brightness of the surface features as they come in and out of view. The rotation periods of such objects can be estimated using the periods of these kinds of variability. Such objects have rotation periods of a few hours to days (Herbig 1954; Cody et al. 2014). In the last few decades, the variability of young stars has been observed over a broad range of wavelengths (Carpenter et al. 2001; Herbst et al. 2002; Rockenfeller et al. 2006; Rebull et al. 2014; Venuti et al. 2015; Ghosh et al. 2021 and references therein).

Some young BDs are surrounded by circumstellar disks, and there is also variability due to unstable accretion from the inner circumstellar disk onto these objects (causing brightening). Recently, a small percentage of rare BDs in young clusters have exhibited more complex rotational modulations (e.g., HSS 348: Strassmeier et al. 2017; Zhan et al. 2019; and EPIC 204117263: Stauffer et al. 2017) and possible poorly known scenarios have been explained by several authors (Stauffer et al. 2017, 2018; Zhan et al. 2019; Günther et al. 2020, 2022; Koen 2021). The atmospherically induced effects could be used to interpret aperiodic and partial periodic variability in BDs. Such variability is more challenging to study than periodic variability, as it has no regular pattern. Thus, the variability can be used to probe several physical mechanisms for these classes of young objects in diverse environments. Therefore, such studies are heightening our knowledge about the dynamic and chemical evolution of the galaxy (Schmidt et al. 2016). Apart from these, short-duration flares are another source of photometric variability and have been frequently observed in active M dwarfs and BDs. Flares are sudden changes in brightness that release a large amount of magnetic energy. Flares can also provide essential information about the strength and nature of the magnetic field (Ramsay & Doyle 2015; Doyle et al. 2022).

At young ages (a few millions of years), the deuterium-burning phase is particularly prominent in BDs, and thus BDs are luminous. Such young objects are detectable through X-rays, UV, and optical/IR wavelengths. Therefore, the young star-forming regions are the best choice for understanding the physical properties and surface features of BDs. The nearby

Taurus star-forming region is an ideal target for probing the surface features of these kinds of objects, as their substellar and planetary-mass members are bright enough to be easily observed by the present all-sky surveys. Young objects are still accreting material from circumstellar disks, which can cause significant variability in their brightness. The greater Taurus (Kenyon et al. 2008) region has been intensively highlighted in many previous photometric observations of young objects in the last few years (Paudel et al. 2018; Rebull et al. 2020; Roggero et al. 2021; Cody et al. 2022; Robinson et al. 2022). Hence, the Taurus region is known to host a large population of young stars, making it an ideal laboratory for studying the photometric variability of young BDs.

The availability of large-time-series photometric data sets at high cadence and high precision from space missions, such as CoRoT (Baglin & Chaintreuil 2006), MOST (Matthews et al. 2000), and the Kepler extended mission K2 (Howell et al. 2014), provides huge scope for understanding the shapes of the light curves of young objects, their rotation periods, and their surface features in the photosphere and dynamical atmosphere. In addition, Cody et al. (2014) have analyzed a large sample of disk-bearing stars in NGC 2264 with high-cadence photometric time-series data from CoRoT and Spitzer in the mid-IR. The Upper Scorpius and ρ Ophiuchus regions were then observed in K2. They provided the optical light curves of young cluster members (Cody et al. 2017; Ansdell et al. 2018; Cody & Hillenbrand 2018; Rebull et al. 2018). Several similar kinds of studies have been performed using ground-based telescopes to probe the characteristics of BDs in young star-forming regions, e.g., Taurus (Luhman et al. 2009; Esplin & Luhman 2017), IC348 (Ghosh et al. 2021), σ Ori (Caballero et al. 2004), and Upper Scorpius (Cody & Hillenbrand 2018; Lodieu et al. 2018). In recent years, the Transiting Exoplanet Survey Satellite (TESS; Ricker et al. 2015) has also provided high-cadence, high-precision light curves of young objects, covering many young clusters, including the extended Taurus star-forming region. TESS can detect even fainter objects than K2 and CoRoT had observed earlier. Time-series data from the all-sky TESS mission provide more opportunities to understand the rotation periods, the amplitudes of starspot modulation, which provide better visualizations of surface features and atmospheric properties, and the energies of optical flare events (e.g., Davenport et al. 2014; Hawley et al. 2014; Stelzer et al. 2016; Ilin et al. 2019; Günther et al. 2020; Medina et al. 2020; Raetz et al. 2020; Stelzer et al. 2022). Although the primary goal of TESS is to search for transiting exoplanets around bright stars, it has also revealed several astrophysical phenomena of M dwarfs and BDs, like the shapes of light curves, e.g., the scallop shell structure, complex modulations (Zhan et al. 2019; Günther et al. 2022), stellar periodicities (Medina et al. 2020), and stellar flares (Doyle et al. 2019, 2022).

Here, we report photometric variability studies of a sample of young Taurus BDs based on the TESS all-sky survey. We have used a target list of 11 young BDs (in Table 1), which were spectroscopically well-characterized samples of the spectral type later than M6.0 and confirmed members of the young (∼2–3 Myr) Taurus star-forming region (Briceño et al. 1998; Luhman 2004, 2006; Guieu et al. 2007a). Luhman (2004) established that the spectral type beyond M6 belongs to the BD mass range at the Taurus age of 2–3 Myr. This Taurus star-forming region has been extensively surveyed and characterized by several authors (Briceño et al. 2002; Luhman 2006; Guieu et al. 2007a; Luhman et al. 2009, 2010, 2017; Rebull et al. 2010, 2011; Esplin et al. 2014; Zhang et al. 2018). The present work aims to probe the dynamic atmosphere and surface features and measure the stellar rotation periods of such young objects. Here, we have reported the rotation periods of BDs in Taurus using the TESS light curves and period analysis. We have also analyzed the flared light curves to estimate the flare energy and interpret their magnetic activities.

This paper describes the sample selection in Section 2. Section 3 describes the TESS observations and data analysis. In Section 4, we discuss the individual objects and obtained results from the studies. Section 5 describes the identification of stellar flares and the results from the flare analysis. Finally, we present a discussion in Section 6, and we have summarized our work in Section 7.

2. Sample Selection

This paper is based on TESS observations of a sample of 11 young (2–3 Myr) BDs in the spectral range from M6 to M8, which have been spectroscopically well characterized and are confirmed members of the Taurus star-forming region (Briceño et al. 1998; Luhman 2004, 2006; Guieu et al. 2007a). To reconfirm the membership of these targets in the Taurus association, we have also used the BANYAN Σ Bayesian analysis tool[1] (Gagné et al. 2018). BANYAN Σ determines the membership probability of astronomical objects in all 27 known and well-characterized young associations within 150 pc, and it shows that all the 11 members have more significant than 95% membership probability in Taurus. This tool calculates a Bayesian membership probability from an object's sky position, proper motion, distance, and radial velocity values. The Taurus star-forming region is a well-studied region because of its proximity ($d = 140$ pc; Torres et al. 2012; Esplin & Luhman 2017), and it also has a relatively large stellar population of low-mass stars, including several BDs ($N \sim 500$; Kenyon et al. 2008), making it an ideal laboratory for studies of BDs. The effective temperatures of these BDs are in the range from 2724 K to 3339 K (Paegert et al. 2021). Details of those BDs are given in Table 1.

3. TESS Observations and Data Analysis

The TESS satellite started its journey into orbit on 2018 April 18, with the primary goal of searching for transiting exoplanets around the brighter stars across the entire sky. TESS has four telescopes, of size 10.5 cm, with wide-field cameras that continuously observe a 24×96 deg^2 strip in the sky for 27 days consecutively (see Ricker et al. 2014 for details). In our present studies, we have selected a few BDs (in Table 1) in the Taurus star-forming region with a 2 minutes cadence of TESS observations. Given a TESS magnitude larger than 14 mag in the TESS broad band (600–1000 nm), the space telescope observed 11 BDs with cameras 2 and 3 during sectors 43 and 44 from 2021 September 16 to 2021 November 5, with a roughly one-day gap at spacecraft perigee. First, we used the "lightkurve" (Lightkurve Collaboration et al. 2018) package to download the TESS light curves of our targets for sectors 43 and 44 from the Mikulski Archive for Space Telescopes (MAST) archive). Then the Science Processing Operations Center (SPOC) pipeline (Jenkins et al. 2016) extracts the 2

[1] http://www.exoplanetes.umontreal.ca/banyan/

Table 1
The Sample of Young BDs in the Taurus Star-forming Region in This Work

TIC	Other Name	R.A. (hr: min:sec)	Decl. (deg: min:sec)	TESS Mag	SpT	$T_{\rm eff}$ (K)	J	H	K	References
456944264	MHO 4	04:31:24.06	+18:00:21.4	14.27	M7.0	2550	11.65	10.92	10.57	B02
150005829	CFHT-BD-Tau 6	04:39:03.96	+25:44:26.3	15.42	M7.25	2506	12.65	11.84	11.37	G06
150122702	CFHT-BD-Tau 8	04:41:10.78	+25:55:11.5	16.39	M5.5	2814	13.19	12.12	11.44	L04
268324578	CFHT-BD-TAU 7	04:32:17.87	+24:22:14.9	14.04	M6.5	2638	11.54	10.79	10.38	G06
58601455	CFHT-BD-Tau 9	04:24:26.46	+26:49;50.2	15.27	M6.25	2682	12.88	12.19	11.76	M10
118520413	CFHT-BD-Tau 11	04:35:08.51	+23:11:39.9	14.72	M6.75	2595	12.53	11.94	11.59	G06
17308621	2MASS J04221332 +1934392	04:22:13.32	+19:34:39.3	15.57	M8.0	2375	12.86	12.05	11.52	L06
125882806	2MASS J04380083 +2558572	04:38:00.87	+25:58:57.2	13.29	M7.25	2507	11.54	10.62	10.10	L10
125882890	2MASS J04381486 +2611399	04:38:14.89	+26:11:39.9	17.84	M7.25	2507	15.18	14.13	12.98	L04
125977598	2MASS J04442713 +2512164	04:44:27.14	+25:12:16.4	13.84	M7.25	2507	12.19	11.36	10.76	L04
56624878	2MASS J04141188 +2811535	04:14:11.88	+28:11:53.4	15.82	M6.25	2682	13.16	12.33	11.64	B02

Note. Column (1): TIC ID of the sources. Column (2): Other IDs. Column (3): R.A. Column (4): Decl. Column (5): TESS magnitudes in the broad band (600–1000 nm). Column (6): Spectral type. Column (7): Effective temperature, using the linear relation, i.e., $T_{\rm eff} = 3780–175.6 \times SpT$ (Guieu et al. 2007a). Columns (8)–(10): magnitudes in the *J, H,* and *K* bands (Cutri et al. 2003). Column (11): References used for spectral type: B02 = Briceño et al. (2002); G06 = Guieu et al. (2007a); L04 = Luhman (2004); L06 = Luhman (2006); L10 = Luhman et al. (2010); and M10 = Monin et al. (2010).

minutes cadence data. Finally, we analyzed the light curves from each sector individually. For our analysis, we have chosen to use the Presearch Data Conditioning (PDCSAP) light curves in our analysis (Smith et al. 2012; Stumpe et al. 2012), which are Simple Aperture Photometry values (SAP FLUX) after removing the long-term trends using Cotrending Basis Vectors. The "presearch data conditioning" (PDC) module also recognizes and records outliers and makes corrections for the crowding contamination from surrounding nearby stars, which the SAP FLUX does not (Guerrero et al. 2021). We have taken only the data with QUALITY = 0, while a nonzero value of QUALITY indicates that the data have been compromised to some degree by instrumental effects. We also removed the outliers and NaNs and normalized each light curve by dividing each point's flux by the target's mean flux.

3.1. Period Analysis of the TESS Light Curves

All 11 light curves of the BDs were inspected visually first, and it was found that more than half of the sample showed clear signs of periodic variability. The objects of our sample are listed in Table 1, including their Tess Input Catalog (TIC) ID, other IDs, R.A., decl., TESS magnitude, spectral type, effective temperature ($T_{\rm eff}$), and magnitudes in the *J, H,* and *K* bands (Cutri et al. 2003; Guieu et al. 2007a, 2007b; Stassun et al. 2018). The effective temperature was calculated from the spectral type using a linear relationship, i.e., $T_{\rm eff} = 3780–175.6 \times$ SpT (Guieu et al. 2007a). To search for a significant periodic signal of each object, we used a Lomb–Scargle (LS) periodogram using the lightkurve package (Lightkurve Collaboration et al. 2018). The LS periodogram (Lomb 1976; Scargle 1982) method uses a Fourier power spectrum estimator to find out significant periodicity, and it can deal with the observed sparse data sets. We also checked the periods using the NASA EXOPLANET ARCHIVE PERIODOGRAM Service[2] to find the significant periodic signals. The periods were further verified with the publicly available STARLINK[3] software database, which includes the systematic errors. All the periodograms in the three different software programs provide fairly close values of the significant peaks in our analysis. The phase light curves are created using the most significant peaks in the periodograms. The periods estimated using the NASA EXOPLANET ARCHIVE PERIODOGRAM Service and STARLINK software for these objects are listed in Table 2. We found that most of the phase-folded light curves are sinusoidal variables. As their periods are typically explicit, false alarm probability (FAP) levels of the significant peaks are either close to 0 or $<10^{-4}$. In addition, the signatures of flare activities were also observed in a few BDs, which are analyzed separately in the later Section 5. We have discussed the details of the individual targets below.

4. Individual Objects and Results

TIC 456944264 (MHO 4). TIC 456944264 (MHO 4) was discovered by Briceño et al. (1998), showing a strong Li I λ6707 absorption line along with He I λ5876, [O I] λ6300, and [O I] λ6363, confirming a late M-type object. MHO 4 is a young BD with a spectral type of M7.0 (Briceño et al. 2002). X-ray emission was detected from this young object in ROSAT and XMM-Newton observations (Carkner et al. 1996; Neuhäuser et al. 1999). The distance to MHO 4 is 144 pc, estimated from GAIA by Bailer-Jones et al. (2018). The observed light curves in the TESS 43 and 44 sectors were analyzed separately, and we detected periods of 2.211 days for sector 43 and 2.269 days for sector 44 using the LS periodogram analysis. Rebull et al. (2020) reported the period of MHO 4 as around 2.2098 days using K2 observations, while Güdel et al. (2007) reported it as <6.29 days using the XMM-Newton X-ray observatory. So, the TESS observation clearly matches well with Rebull et al. (2020), and it also falls within the upper limit mentioned in Güdel et al. (2007). Flaring events in MHO 4 are also observed in each sector of TESS observations. For example, the PDCSAP FLUX light

[2] https://exoplanetarchive.ipac.caltech.edu/

[3] http://starlink.eao.hawaii.edu/starlink

Table 2
New Results of Periods from TESS Data

TIC	Periods in This Work (days)				Previous Periods (days)				
	Method 1		Method 2		Cody et al. (2022)	Rebull et al. (2020)	Scholz et al. (2018)	Güdel et al. (2007)	Roggero et al. (2021)
	TESS Sector 43	TESS Sector 44	TESS Sector 43	TESS Sector 44					
456944264	2.211	2.269	2.211 ± 0.050	2.286 ± 0.054	…	2.2098	…	6.29	…
150005829	3.126	3.132	3.085 ± 0.100	3.109 ± 0.100	3.12	…	3.3	3.1401	…
150122702	3.607	3.263	3.545 ± 0.132	3.127 ± 0.107	3.91	…	…	…	3.91
268324578	…	…	…	…	1.72, 2.60	…	…	…	…
58601455	2.403	7.113	2.426 ± 0.061	6.874 ± 0.493	…	…	…	…	…
118520413	1.495	1.496	1.494 ± 0.023	1.497 ± 0.023	…	1.4981	1.5	…	…
17308621	1.330	1.331	1.337 ± 0.018	1.338 ± 0.019	1.33	…	…	…	…
125882806	1.025	1.029	1.028 ± 0.011	1.025 ± 0.011	0.66	0.6644, 1.0263	2.01	…	…
125882890	…	…	…	…	…	…	…	…	…
125977598	4.444	4.429	4.362 ± 0.199	4.393 ± 0.200	4.46	4.43	4.48	…	4.42
56624878	…	…	…	…	…	…	…	…	…

Note. Method 1 = periods determined using the NASA EXOPLANET ARCHIVE PERIODOGRAM SERVICE. Method 2 = periods computed using the STARLINK software.

curve, the LS periodogram, and the phase light curve are shown in Figure 1 for sectors 43 and 44, respectively.

TIC 150005829 (CT 6). TIC 150005829 (CFHT-BD Tau 6; hereafter, CT 6) has an M7.25 spectral type, confirmed by Luhman (2004). X-ray emission in CT 6 has also been detected, and it was interpreted as an accreting BD (Muzerolle et al. 2005; Grosso et al. 2006), with an estimated accretion rate of $\log_{10} \dot{M} (M_\odot \mathrm{yr}^{-1}) = -11.3$. The TESS observations in sectors 43 and 44 show significant variation in their light curves. We estimated a rotation period of around 3.13 days in both sectors. The phase light curves of sectors 43 and 44 are shown in Figure 1. Rebull et al. (2020) have determined the period of CT 6 as 3.14 days using K2 observations, while Scholz et al. (2018) have estimated a period of around 3.19 days from the K2SFF light curve, using the autocorrelation function, and 3.33 days, using the Gaussian Process method. Again, using K2 data, Cody et al. (2022) estimated a period of 3.12 days, and classified it as a quasiperiodic symmetric variable object. So, our estimated period for CT 6 from the TESS observations matches well with the previously reported values of the periods.

TIC 150122702 (CT 8). TIC 150122702 (CFHT-BD Tau 8; hereafter, CT 8) is a spectroscopically confirmed object of spectral type M5.5, by Luhman (2004), while Guieu et al. (2007a) derived an M6.5 spectral type, with reddening $Av = 1.8$. Guieu et al. (2007a) also found that it exhibits a level of Hα emission in excess of chromospheric activity. From K2 observations, Roggero et al. (2021) estimated a period of 3.91 days, while Cody et al. (2022) reported a quasiperiodic dipper with the same period. The TESS observations show a similar quasiperiodic dipper pattern in the light curves of both sectors. The largest peak in the LS periodogram is at 3.607 days in TESS sector 43, while that of 3.263 days is found in TESS sector 44. Folding the light curves at those periods provides a gross sinusoidal variation with angular dips in their phase light curves (Figure 1).

TIC 268324578 (CT 7). TIC 268324578 (CFHT-BD-Tau 7; hereafter, CT 7) is a recently discovered M6.5 dwarf Guieu et al. (2007a). Guieu et al. (2007a) proposed that CT 7 is possibly associated with V928 Tau (SpT M0.5) at ≈ 2250 au. Furthermore, Guieu et al. (2007b) found that it has a weak Hα emission, but no excess in IR emission. While examining the target pixel file (TPF) of CT 7 with the `interact_sky` tool, we identified the presence of a nearby bright object within the SPOC pipeline aperture. The nearby source (TIC 268324580) is approximately three TESS magnitudes brighter than CT 7. This object is located only 18″ from CT 7. Notably, upon comparing the light curves of both objects, we found them to be identical. Based on this contamination effect, we excluded CT 7 from our study.

TIC 58601455 (CT 9). TIC 58601455 (CFHT-BD-Tau 9; hereafter, CT 9) has been classified as an M6.25 dwarf (Monin et al. 2010), and the authors also reported the presence of a disk around this object from the mid-IR data of Spitzer. For TESS sector 43, we estimate only a single strong peak at 2.40 days in the LS periodogram, while in TESS sector 44, it shows three strong resolved peaks, with $P_1 = 4.589$ days, $P_2 = 7.113$ days, and $P_3 = 12.029$ days. Among these, P_2 is the most significant peak. The power at P_2 is 1.86 times higher than at P_1. CT 9 in TESS sector 43 does not show any significant evolution, with a folding period of 2.40 days (Figure 1), while in TESS sector 44 (Figure 1), we find a sinusoidal shape after phasing the light curve to all periods.

TIC 118520413 (CT 11). TIC 118520413 (CFHT-BD-Tau 11; hereafter, CT 11) is an M 6.75 dwarf and a confirmed member of Taurus (Guieu et al. 2007a). The authors also detected excess Hα emission with equivalent width (EW) ≈ 45.07 Å, which indicates the chromospheric activity. The LS periodogram shows a dominant, strong peak at a short period of 1.495 days in TESS sector 43 and 1.496 days in sector 44. When phased to those periods, the light curve has a sinusoidal appearance in both sectors (Figure 1). Scholz et al. (2018) and Rebull et al. (2020) reported the periods as 1.5 days and 1.4981 days, respectively, using K2 observations, which match well with the TESS results. We have also identified one flare event for CT 11 during the TESS observations in sector 43. The details of the flare energy calculation are described in Section 5.1.

TIC 17308621 (2M J0422+1934). TIC 17308621 (2MASS J04221332+1934392; hereafter, 2M J0422+1934) was first discovered by Luhman (2006) and classified as a spectral type of an M 8.0 dwarf. The author revealed that it is a confirmed

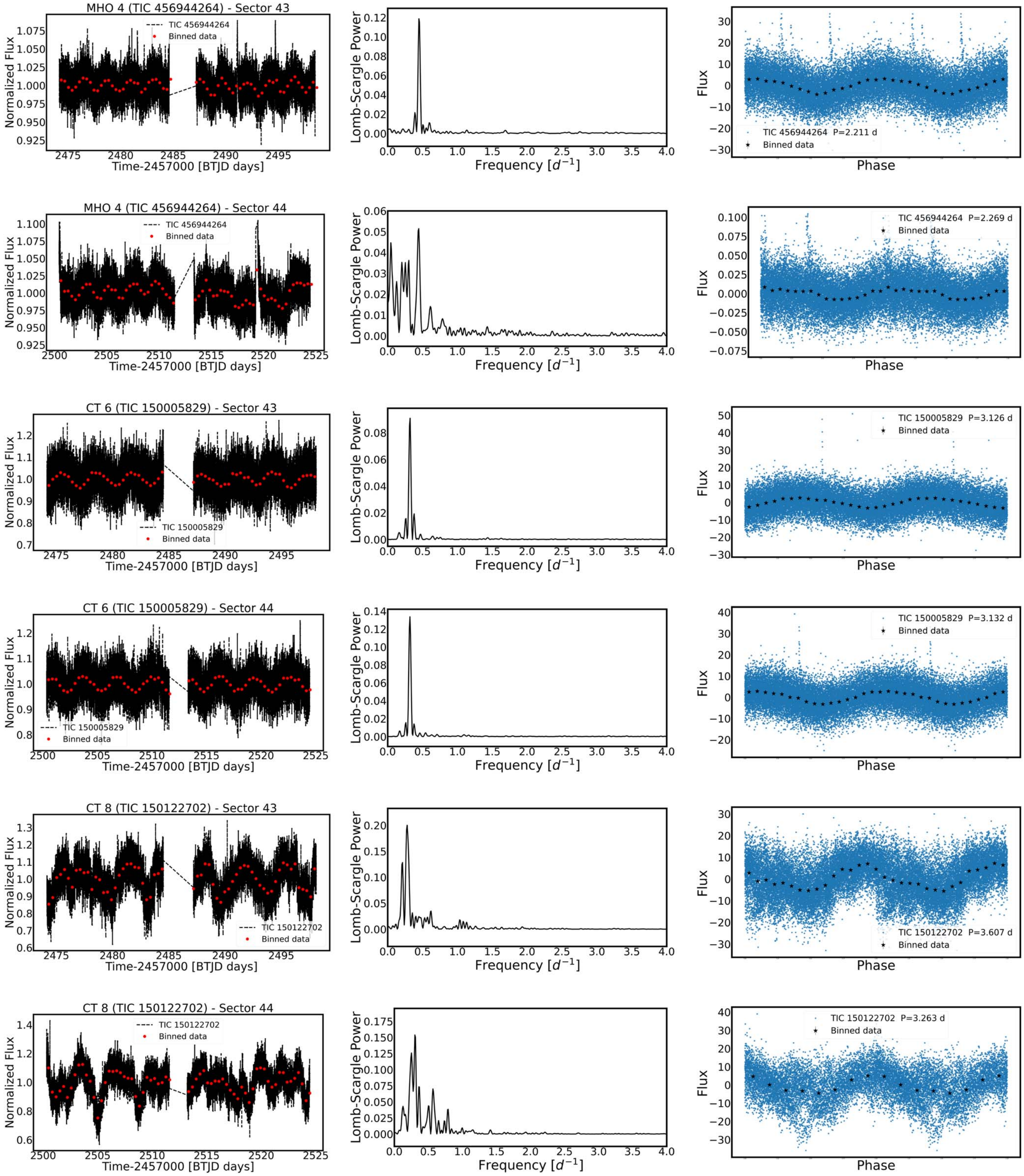

Figure 1. The light curves of all selected BDs in the Taurus star-forming region are shown here. The left column showcases the complete light curves, displayed in black, while the binned light curves with a binning interval of 500 minutes are depicted as red dots. In the middle column, the LS periodograms are showcased, illustrating the power spectra of the BDs' rotational periods. Finally, the right column displays the phase-folded light curves, where the data are folded with the most significant peak obtained from the LS periodogram. The black stars represent the 1000-point binned data. The object names, TIC IDs, and sectors are included in the figure titles in the left column. Additionally, the rotation periods of the BDs are mentioned within the phase light curves.

member of Taurus, with alkali lines NaK as membership evidence. This source exhibits X-ray emission detected by Grosso et al. (2006). Güdel et al. (2007) found no evidence of X-ray variability with the XEST survey. The TESS observations in sectors 43 and 44 show clear variability in the optical light curves. From both sectors' light curves of 2M J0422+1934, we estimate a strong dominant peak around 1.33 days in the power spectrum. After phasing the light curve

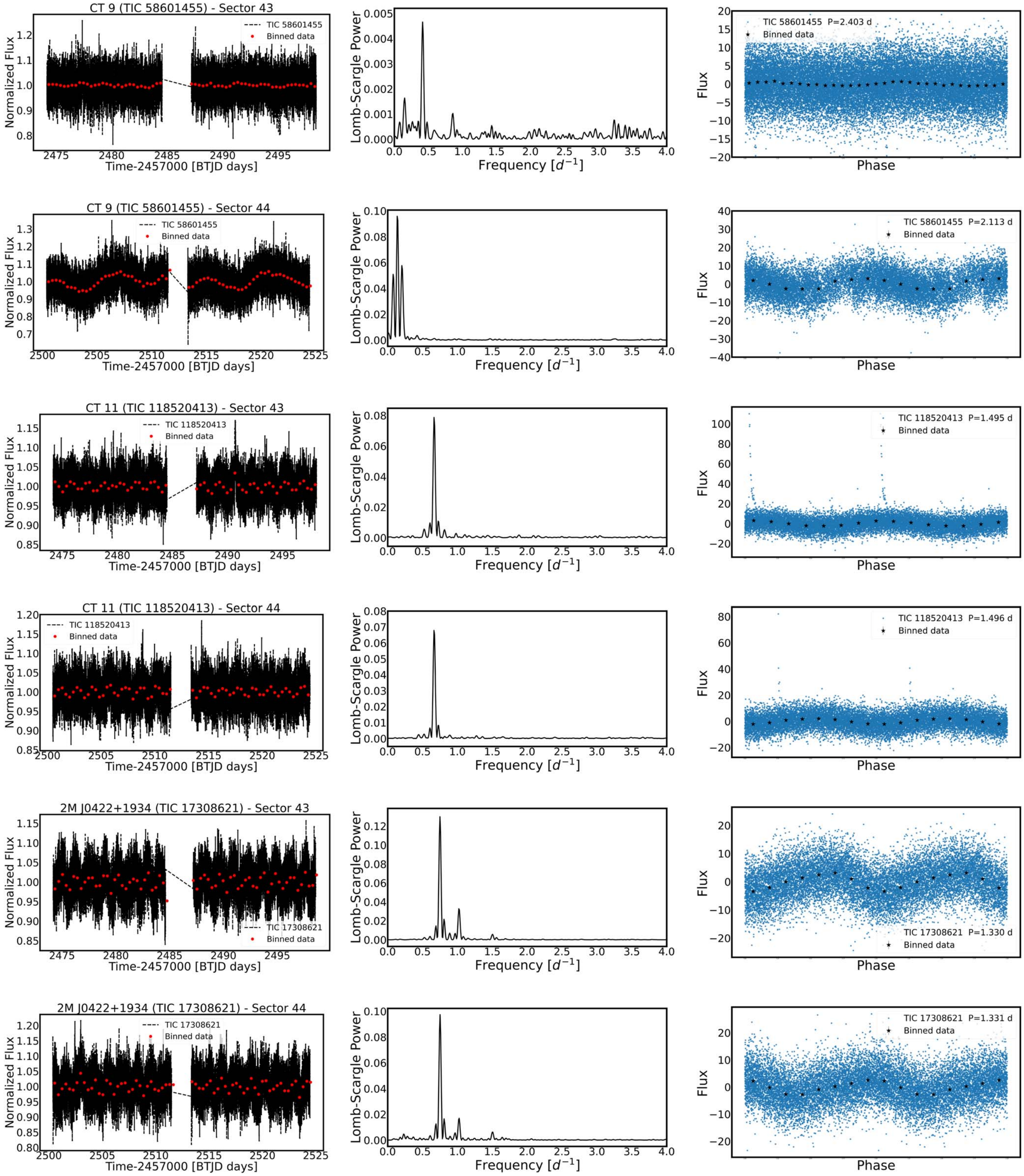

Figure 1. (Continued.)

to that period, we find a well-approximated sinusoidal pattern (Figure 1). Cody et al. (2022) also reported the same period from K2 data.

TIC 125882806 (2M J0438+2558). TIC 125882806 (2MASS J04380083+2558572; hereafter, 2M J0438+2558) has spectral type of M 7.25 (Luhman et al. 2010). 2M J0438+2558 was first discovered by Luhman (2004), with confirmed membership evidence of NaK. Previously, from K2 observations, Scholz et al. (2018) reported a period of 2.01 days, while Rebull et al. (2020) detected two periods of 0.6644 days and 1.0263 days for this object from K2 data. Cody et al. (2022) recently reported that it has multiple distinct periods, with a primary peak around 0.66 days, with K2 data. Our periodogram analysis of the TESS observations estimates only one robust

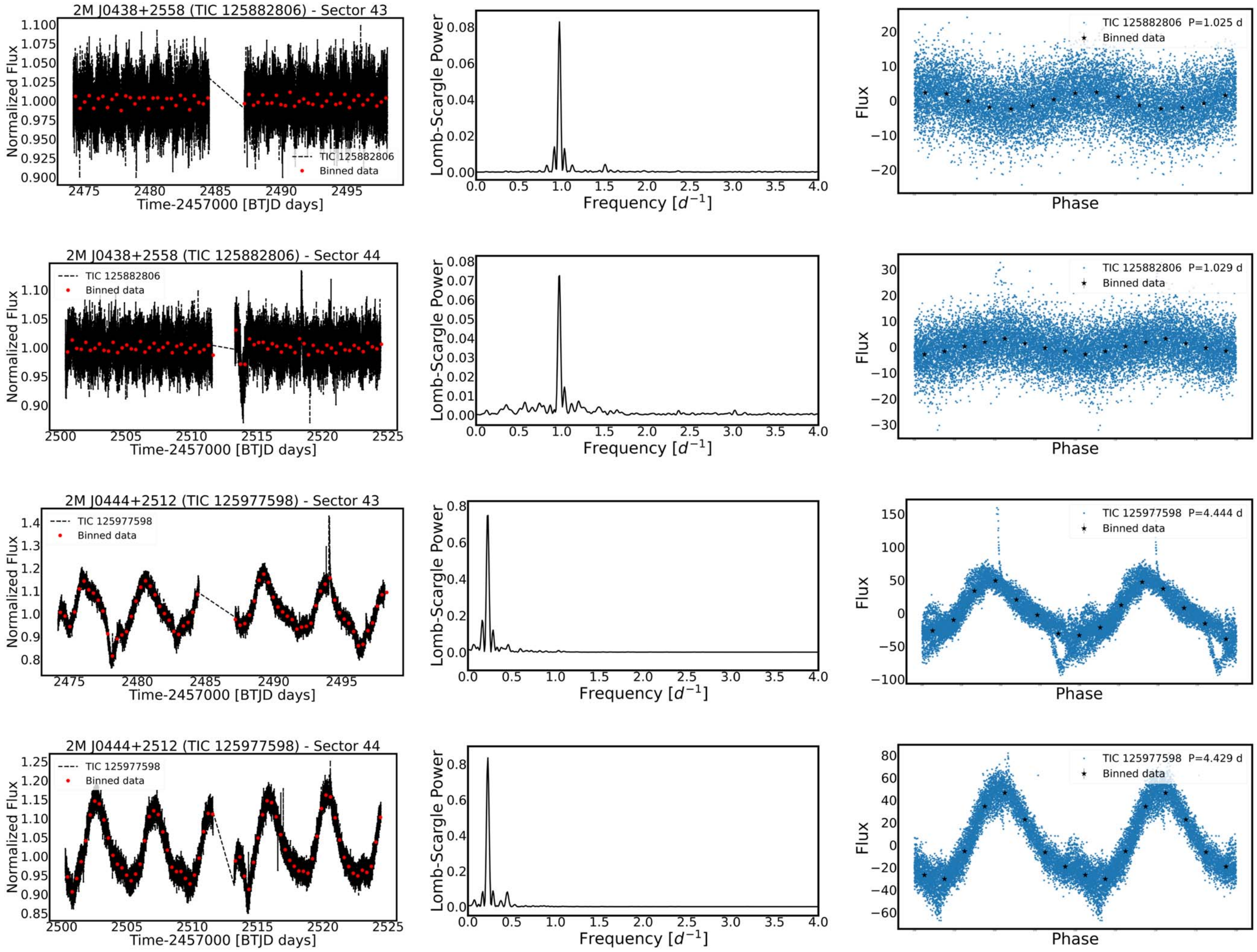

Figure 1. (Continued.)

and real peak, at 1.025 days and 1.029 days, in sectors 43 and 44, respectively. The second period in Rebull et al. (2020) matches the period we obtained from the TESS data. Our estimated period is about half of the period obtained by Scholz et al. (2018), maybe a harmonic. When we phased the light curve to this period, we found clear periodic variation in both sectors (Figure 1).

TIC 125882890 (2M J0438+2611). TIC 125882890 (2MASS J04381486+2611399; hereafter, 2M J0438+2611) has been classified as an M 7.25 spectral type with mass $70M_{\rm Jup}$ (Luhman 2004; Muzerolle et al. 2005), which is located in a dense region of the Taurus molecular cloud and is close to GM Tau, with a rough distance $\approx 2\farcm8$ (Goldsmith et al. 2008; Phan-Bao et al. 2014). Luhman (2004) reported that it shows strong forbidden emission lines, suggesting an outflow activity. It also exhibits mid-IR excess emission that indicates the presence of a disk (Luhman 2006). Luhman et al. (2007) proposed that it has a transition disk with an inner radius $\approx$0.275 au. 2M J0438+2611 is also reported as a strong accreting BD, with accretion rate $\log_{10}\dot{M}\,(M_\odot yr^{-1}) \approx -10.8$ and EW (Hα) $\approx$ 47 Å (Muzerolle et al. 2005). The object shows a flat light curve in both sectors of TESS. From the periodogram analysis, we have checked that no significant peaks are apparent for this object in either sector. All the peaks are below the FAP. We have not included the light curves of this object. However, from K2 observations, Cody et al. (2022) classified its variability as a burster, as similarly detected for EPIC 211029507 (DH045) by Rebull et al. (2016a).

TIC 125977598 (2M J0444+2512). TIC 125977598 (2MASS J04442713+2512164; hereafter, 2M J0444+2512) is known as an M 7.25 young BD, initially identified with IRAS far-IR data by Knapp et al. (2004). Luhman (2004) estimated an effective temperature of 2838 K, a luminosity of $0.028L_\odot$, and a strong Hα emission, with EW (Hα) = 100 Å. Bouy et al. (2008) estimated an accreting rate of $\log_{10}\dot{M}\,(M_\odot yr^{-1}) = -9.7$. The disk around 2M J0444+2512 was first inferred from IR data (Kenyon et al. 1994; Luhman 2006; Hartmann et al. 2005; Guieu et al. 2007b). From high-angular-resolution observations with CARMA at 1.3 mm, a spatially resolved dusty disk around this BD was detected for the first time. There is evidence of larger-sized (on the order of millimeter) grains in the disk (Ricci et al. 2013). Ricci et al. (2017) revealed a radio emission from this BD using Very Large Array (VLA) observations and reported that the disk around this BD might be the most massive. From K2 observations, Roggero et al. (2021) estimated a rotation period of 4.42 days with a dipper in the phase light curve, similar to the reported periods of 4.48 days by Scholz et al. (2018) and 4.43 days by Rebull et al. (2020). Furthermore, using K2 data, Cody et al. (2022) reported that it is a quasiperiodic symmetric

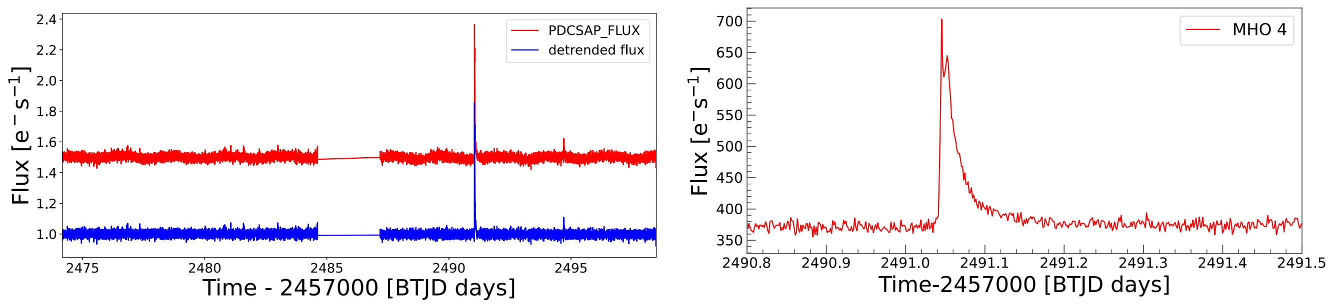

Figure 2. The left panel shows the PDCSAP light curve (red) and the detrended light curve (blue) of MHO 4 in TESS sector 43. The right panel shows the zoomed-in view of the flare. The *x*-axis corresponds to the time in Barycentric TESS Julian Days (BTJD), and the *y*-axis corresponds to the TESS flux (e^{-1}/s).

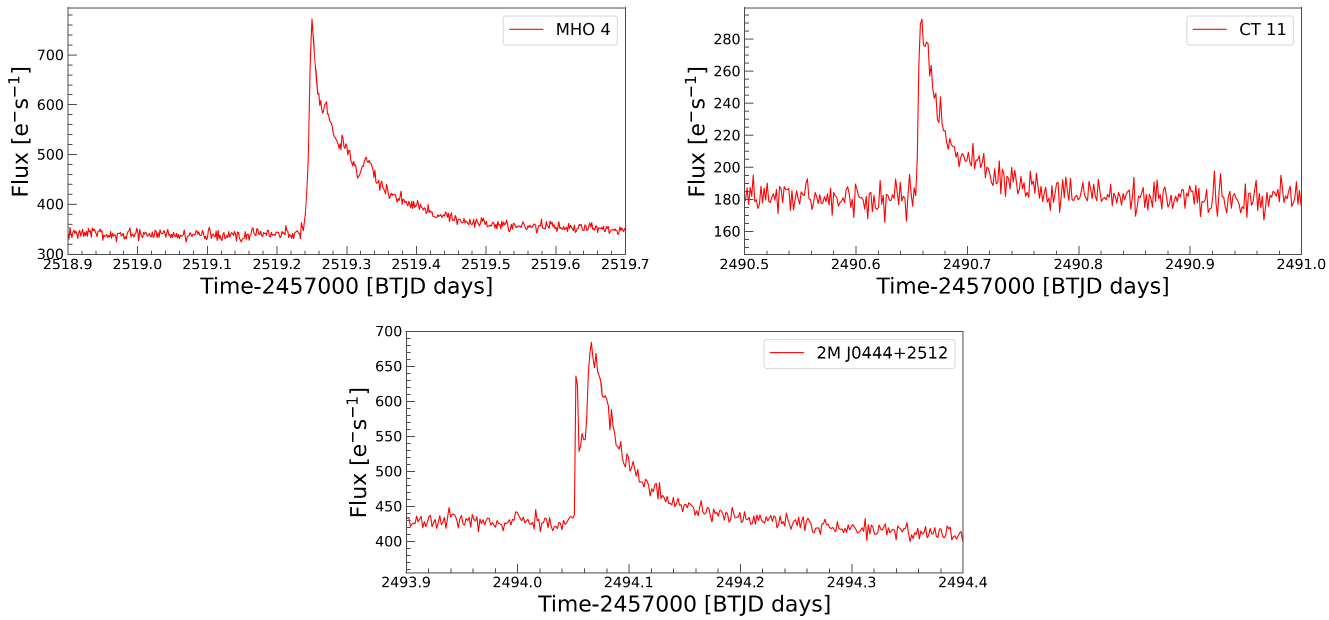

Figure 3. Flare events for MHO 4 in sector 44 (top left), CT 11 (top right), and 2M J0444+2512 (below).

variable with a period of 4.46 days. From our analysis of the TESS data, 2M J0444+2512 has also shown the largest peaks in the LS periodogram, at 4.444 days and 4.429 days in sector 43 and sector 44, respectively. The phase-folded light curve (Figure 1) shows a sinusoidal structure with a dipper pattern, which is well matched with the previous results from K2. This BD shows a strong flare in TESS sector 43, analyzed in the later Section 5.1.

TIC 56624878 (2M J0414+2811). TIC 56624878 (2MASS J04141188+2811535; hereafter, 2M J0414+2811) was spectroscopically confirmed as a young BD in the Taurus molecular cloud and classified as an M 6.25 dwarf (Briceño et al. 2002; Luhman 2004). 2M J0414+2811 shows a strong Hα emission, with EW $\approx$250 Å, with the accretion rate $\log_{10}\dot{M}\,(M_\odot\,\mathrm{yr}^{-1}) \approx -10$. It also exhibits a strong excess of the continuum at short wavelengths, which indicates high accretion from the circumstellar disk (Luhman 2004; Muzerolle et al. 2005). But Grosso et al. (2007) reported that it is only 24″ away from the weak-line T Tauri star V773 Tau. To verify it, we used the `interact_sky` tool in the TPF of 2M J0414+2811 and confirmed that it is indeed contaminated by this nearby bright source. However, upon comparing the light curves of both sources, we observed that they were fairly identical. Considering the contamination and similarity of the light curves, we excluded 2M J0414+2811 from this study.

5. Detection of Stellar Flares

We have identified four flaring events in the TESS light curves from three young BDs, i.e., MHO 4, CT 11, and 2M J0444+2512 in TESS sectors 43 and 44. For MHO 4, we have detected two flares, one in each sector. For CT 11 and 2M J0444+2512, we have detected only one flare event, in sector 43. For example, we have shown the flare and detrended light curve of MHO 4 for sector 43 in Figure 2, while Figure 3 illustrates zoomed-in views of the flare events from MHO 4 (in sector 44), CT 11, and 2MASS J0444+2512.

5.1. Computation of Flare Energies

To calculate the energy of a flare, we first estimated its equivalent duration (ED), which is the equivalent time during when a substellar object (in its quiescent state) would have emitted the same amount of energy as the flare emitted (Gershberg 1972). It is the area under the flare light curve in the unit of time. It depends on the filter and photospheric properties of the objects, but not on the distance to the objects (Paudel et al. 2019). To calculate the ED, we implemented the open-source Python software ALTAIPONY[4] (Ilin et al. 2021), which automatically detects and characterizes the flares in our sample. We used the PDCSAP light curve and detrended the light

[4] https://altaipony.readthedocs.io/en/latest/

Table 3
The Details of the Flare Properties of the Three Young BDs in the TESS Observations

Object	Sector	Flares	Radius ($R_\odot$)	Start Time (BTJD-2457000)	Stop Time (BTJD-2457000)	Amplitude	ED (sec)	ED Err (sec)	Energy (erg)	Duration (min)
MHO 4	43	1	0.655	2491.042674	2491.101013	0.857	1479.792	11.104	2.26E+36	84
MHO 4	44	1	0.655	2519.240429	2519.447383	1.209	5852.146	22.173	8.94E+36	298
CT 11	43	1	0.440	2490.655007	2490.698066	0.598	1046.460	20.949	7.31E+35	62
2M J0444+2512	43	1	0.607	2494.052309	2494.125926	0.632	1821.460	14.540	3.34E+36	106

Note. This table includes the number of flares in each sector, the stellar radius from the TIC (Stassun et al. 2018), the flare start time and stop time, the amplitude of the flare in the light curve, the ED and its uncertainty, the energy of the flare, and the duration time.

curves to remove rotational modulation trends. We applied the Savitzky–Golay filter (the "savgol" package; Savitzky & Golay 1964) to the light curves for detrending. The "savgol" package is computationally fast and gives reliable results for the TESS light curves. These flares were identified by the *find_flares()* package in the detrended light curves, detecting those points as flares that lie 2.5σ higher than the mean (see Doyle et al. 2018, 2019 for more details). Using the ALTAIPONY analysis gives several flare parameters, i.e., the start and stop time, flare amplitude, and ED. The duration of the flare was calculated using the start and stop time, and the ED was used to calculate the energy of flares. In order to calculate the flare energy $E_{\rm flare}$, we generated synthetic photospheric spectra of three flaring objects by using the BT-Settl model.[5] The parameters—like the effective temperatures, surface gravities, and metallicities—of the respective objects used to generate the synthetic spectra were taken from Stassun et al. (2018). We assumed that the observed flare had a temperature $T_{\rm flare}$ of 9000 K (Hawley & Fisher 1992). Kretzschmar (2011) found that the blackbody spectrum's spectral energy distribution of flares can be fitted well with that temperature. Therefore, the bolometric flare luminosity $L_{\rm flare}$ can be estimated using that temperature $T_{\rm flare}$ and the area $A_{\rm flare}$ of the flare from the following equation of Shibayama et al. (2013):

$$L_{\rm flare} = \sigma_{\rm SB} T^4_{\rm flare} A_{\rm flare}, \tag{1}$$

where $\sigma_{\rm SB}$ is the Stefan–Boltzmann constant. The flare amplitude is defined as $\Delta F/F = (F_i\text{-}F_o)/F_o$, where F_i is the maximum flux at the flare and F_o is the local mean flux (Hawley et al. 2014). Using the flare amplitude ($C_{\rm flare}$), the observed luminosity of the star ($L'_{\rm star}$), and the observed luminosity of the flare ($L'_{\rm flare}$), we estimate $A_{\rm flare}$:

$$L'_{\rm star} = \int R_\lambda F_{\lambda(T_{\rm eff}))} d\lambda . \pi R^2_{\rm star}, \tag{2}$$

$$L'_{\rm flare} = \int R_\lambda B_{\lambda(T_{\rm flare})} d\lambda . A_{\rm flare}, \tag{3}$$

$$C_{\rm flare} = \frac{L'_{\rm flare}}{L'_{\rm star}} = \frac{(F_i - F_o)}{F_o}, \tag{4}$$

where λ is the wavelength, $B_{\lambda(T)}$ is the Planck function, $F_{\lambda(T)}$ is the flux of the synthetic photosphere spectrum, and R_λ is the response function of the TESS bandpass. Therefore,

$$A_{\rm flare} = C_{\rm flare} \pi R^2_{\rm star} \frac{\int R_\lambda F_{\lambda(T_{\rm eff})} d\lambda}{\int R_\lambda B_{\lambda(T_{\rm flare})} d\lambda}. \tag{5}$$

Then the total bolometric energy of the flare ($E_{\rm flare}$) is an integral of $L_{\rm flare}$ during the flare duration (Shibayama et al. 2013):

$$E_{\rm flare} = \int L_{\rm flare}(t) dt. \tag{6}$$

Using Equations (1), (5), and (6), we get

$$E_{\rm flare} = K \cdot (ED), \tag{7}$$

where K is a constant that depends on R_λ, B_λ, and $T_{\rm flare}$ and ED corresponds to the time-integrated factor of the flare amplitude ($C_{\rm flare}$), which is calculated from the start to the end time of the flare (Hawley et al. 2014; Ikuta et al. 2023):

$$K = \sigma_{\rm SB} \pi R^2_{\rm star} T^4_{\rm flare} \frac{\int R_\lambda F_{\lambda(T_{\rm eff})} d\lambda}{\int R_\lambda B_{\lambda(T_{\rm flare})} d\lambda}. \tag{8}$$

From Equation (7), we have calculated the flared energy. Overall, we identified four flaring events across three BDs in our sample, with energies between 7.31×10^{35} erg and 8.94×10^{36} erg. Table 3 lists the full details of each star's flare properties.

6. Discussion

Inspecting the power spectra from TESS time-series data of Taurus, we found that most of the selected sources in our studies show strictly periodic variables, with low to high amplitudes. Out of the eight variable sources, we found that five sources have just one significant period in the LS periodograms and show a sinusoidal periodic nature by phasing the light curves with the most significant period. The high peak in the periodogram might arise due to large spots/spot groups rotating in and out of view (Rebull et al. 2016a). Three sources show more than one significant period (multiple peaks) in the power spectrum, and the phased light curves are generated using only the dominant peak. Rebull et al. (2020) also provide the variability statistics of the Taurus association based on K2 observations, finding that about 83% of Taurus members are periodic variables, while 70% have only one period and 30% have at least two periods. In many cases, the shape of the phase-folded light curves can be well approximated as sinusoidal. We interpret such variability as being due to cool spots corotating with targets.

In Figure 1 of sector 43, CT 8 has two peaks in the periodogram, but the sinusoidal nature arises with only the most significant peak. In Figure 1 for sector 44, CT 8 has multiple unresolved peaks. The most significant peak is asymmetric, which confirms it is a kind of "complex peak," implying a latitudinal differential rotation or spots appearing/disappearing or changing over time (Rebull et al. 2016b). Other

[5] http://svo2.cab.inta-csic.es/theory/newov2/

peaks are also significant. We get similar trends when we fold the light curve with these significant peaks in sector 44. However, as we go toward the lower significant peaks (2.75 days, 1.98 days, and 1.76 days) from the largest one, the amplitude of the sinusoidal pattern decreases, correspondingly. Such rotational modulation is believed to be due to the cool star spots at different latitudes (Parks et al. 2014). The phase light curves in both sectors have eclipsing-like deep in flux on top of the sinusoidal variation. The source might be an eclipsing binary (Rebull et al. 2022).

MHO 4 has a single dominant peak in sector 43 (Figure 1), while in sector 44 (Figure 1) it has a complex, structured, broader peak in the power spectra. Although the periods in both sectors are very close together, the power spectra change dramatically over the sector, which could be described by seeing spots at different latitudes in differentially rotating stars or a spot/spots that appear, disappear, or change over the sector (Rebull et al. 2016b). Folding the light curve with other significant peaks does not show any variability, except at the most significant period, 3.29 days, which might be due to its relatively low amplitude.

6.1. Flare Events in BDs

We have detected four flare events in three BDs in our sample during TESS observations in sectors 43 and 44, with flared energy in the range 10^{35}–10^{36} erg, which falls in the superflare category (Shibayama et al. 2013). In contrast to the general trend of the flare light curves in low-mass stars and BDs having a rapid rise and gradual decline, here all the flare light curves are showing a complex multipeak behavior in the decline phase, which has previously been observed in many young objects (Rebull et al. 2016a). Multiple peaks and irregular shapes in the flare light curve indicate a more complex underlying physical process. The study of such a flare can provide important insights into the magnetic activity and energy release mechanism in stars. Flare events are thought to be similar to solar flares (Davenport et al. 2014), and if we consider an analogy between solar flares and those in BDs, we can roughly estimate the lower limit of the magnetic field strength ($B_z^{\max}$) to produce such superflares, following the relation given by Aulanier et al. (2013) and Paudel et al. (2018):

$$E_{\rm flare} = 0.5 \times 10^{32} \left(\frac{B_z^{\max}}{1000G}\right)^2 \left(\frac{L^{bipole}}{50Mm}\right)^3 erg, \tag{9}$$

where $E_{\rm flare}$ is the bolometric flare energy and $L^{\rm bipole}$ is the linear separation between bipoles. Taking $L^{\rm bipole} = \pi R$, where R is the radius (taken from Table 3), gives the maximum distance between a pair of magnetic poles on the surface of an object. Based on this calculation, we obtained the lower limit of the magnetic field strength ($B_z^{\max}$), since we assumed the upper limit of the bipole length. To create such superflares, we estimated that a magnetic field strength of 1.4 kG (sector 43) and 2.8 kG (sector 44) is required for the object MHO 4. Additional evidence of chromospheric activities has been seen from X-ray (Neuhäuser et al. 1999) and Hα (Briceño et al. 1998; Duchêne et al. 2017). We also predicted that 1.4 kG and 1.9 kG magnetic field strengths are needed to produce superflares in CT 11 and 2M0444+2512, respectively. Earlier, the strong magnetic field strengths were reported for the young BDs LSR J1835+3259 (∼5 kG; age ∼22 Myr) and CFHT-BD-Tau 4 (∼13.5 kG; age ∼2–3 Myr; Berdyugina et al. 2017; Paudel et al. 2018). In the case of 2M0444+2512, additional evidence—like radio emission using the VLA (Ricci et al. 2017), a jet and an outflow from high-resolution optical spectroscopy (Bouy et al. 2008), and a massive disk from 1.3 mm observations (Ricci et al. 2013)— confirms that it is also an active magnetic object.

In general, M dwarfs are highly magnetic active objects (Scholz et al. 2009), and superflare events are expected. Such superflare activity depends on the size and distribution of the star spots. A minor disturbance in chromosphere activity can enhance the flare energy to that of the superflare range without any other excitation mechanism (Yang et al. 2017; Ghosh et al. 2021). Furthermore, it is interesting to note that the high-energy particles and photons from such superflare events play an important role in the habitability of nearby planets, if any (Airapetian et al. 2020; Aizawa et al. 2022).

7. Summary and Conclusion

In this paper, we have presented TESS time-series photometry observations of 11 well-characterized young BDs of spectral type later than M6.0, which are bona fide members of the Taurus star-forming region. We summarize the main results here:

1. The TESS PDCSAP light-curve analysis helps us to identify the variability in 11 young BDs of Taurus. Most of the light curves (72%) show sinusoidal variability, and one object, 2MASS J0438+2611, does not show any variability.
2. We searched for the rotation periods of the variable objects by using LS periodogram analysis to find any periods. Out of the 11 young BDs, we found that 72% are periodic in the period range of 1–7 days; among them, three BDs have periods <1.5 days and the period of one object (CT 9) was estimated for the first time.
3. We have detected four flare events in three BDs (e.g., MHO 4, CT 11, and 2MASS J0444+2512), and one object, MHO 4, shows two flare events. We estimated the emitted energies and found the energies to be in the range of 7.31×10^{35} erg–8.94×10^{36} erg, which is the superflare category.
4. Assuming a similarity with solar flares, we have estimated the lower limit of the magnetic field strength required to produce such flare events. We have calculated the required magnetic field strength to a few kilogauss. These superflares significantly impact planets' habitability around M-type dwarfs.
5. About 72% of the samples are periodic variables. The rotation modulation of the stellar flux due to unevenly distributed cool spots or groups of spots on the stellar surface causes periodic variability in such low-mass objects.

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Software: astropy (Astropy Collaboration et al. 2013, 2018), astroquery (Ginsburg et al. 2019), tesscut (Brasseur et al. 2019), lightkurve (Lightkurve Collaboration et al. 2018), AltaiPony (Davenport 2016; Ilin et al. 2021), tpfplotter (Aller et al. 2020), numpy (Harris et al. 2020), matplotlib (Hunter 2007).

Data Availability

All the TESS data used in this paper can be found in MAST: doi:10.17909/s236-ac55.

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